

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275509

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

HIGGINS; Kevin W. et al.

HYDROPONICS SYSTEM

Abstract

Provided herein is a system for hydroponics growing of plants. A hydroponics system described herein includes: a water tank; a pump; a supply manifold; a drain manifold; a plurality of lower support brackets, each lower support bracket having a lower support bracket cup; a plurality of upper support brackets, each upper support bracket having an upper support bracket cup, where water is supplied from the water tank to each upper support bracket cup via the supply manifold, where water is returned to the water tank from the lower support bracket cup via the drain manifold. The self-contained hydroponics system of an example embodiment further includes a hydroponics system tower disposed between a respective pair of an upper support bracket cup and a lower support bracket cup, where water flows from the upper support bracket cup, through the hydroponics system tower, and out of the lower support bracket cup.

Inventors: HIGGINS; Kevin W. (Fort Mill, SC), WALLACE; Darren M. (Fort Mill, SC)

Applicant: AQUATREE GLOBAL, LLC (Charlotte, NC)

Family ID: 87312207

Appl. No.: 19/213521

Filed: May 20, 2025

Related U.S. Application Data

parent US continuation 17818592 20220809 PENDING child US 19213521

Publication Classification

Int. Cl.: A01G31/06 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims priority to U.S. patent application Ser. No. 17/818,592, filed on Aug. 9, 2022, the contents of which are hereby incorporated by reference herein.

TECHNOLOGICAL FIELD

[0002] This application relates generally to agriculture, and more particularly to systems (e.g., apparatuses, or the like) and methods for growing vegetation in both large and small scale hydroponic and/or aquaponic configurations.

BACKGROUND

[0003] Hydroponics and aquaponics are burgeoning fields to provide alternatives to conventional farming, particularly where conventional farming is not possible due to climate and/or suitable land availability. Hydroponics and aquaponics both enable the growing of plants without requiring soil. Aquaponics is a vegetation (e.g., vegetables, herbs, spices, fruits, grasses, flowers, plants, or any other type of edible or non-edible vegetation) production system that combines supporting aquatic organisms (e.g., raising fish, snails, crayfish, prawns, or other like organisms in tanks) in aquaculture system with a hydroponic system (e.g., cultivating plants in water outside of the ground) in a symbiotic environment. The by-products (e.g., excretions and waste) from the animals being raised accumulate in tank, which increases the toxicity of the water in the tanks. The by-products are broken down by nitrogen-fixing bacteria into nitrates and nitrites, and the water is fed to the hydroponic systems where plants use the nitrates and nitrites as nutrients. The water is then recirculated back to the aquaculture system. Plants are grown as in hydroponics systems, with their roots immersed in the nutrient-rich water. These systems working together enable the ammonia that is toxic to the aquatic animals to be filtered out of the system, while at the same time providing nutrients to the plants. After the water has passed through the hydroponic subsystem, it is cleaned and oxygenated, and can return to the aquaculture tanks. This cycle is continuous.

[0004] As existing hydroponic and aquaculture farming techniques form the basis for all aquaponics systems, the size, complexity, and types of vegetation grown in an aquaponics systems can vary as much as any system found in either distinct farming discipline. Not all existing systems of aquaponics can be satisfactorily applied to indoor large and/or small scale use. Issues such as appearance, humidity, where the water is routed if a leak occurs, and adaptability for farming different types of vegetation are all considerations for indoor systems. The present invention addresses these needs by providing an aquaponics system that allows for the production of various types of vegetation, in various scales, and in various locations.

BRIEF SUMMARY

[0005] Embodiments of the invention are directed to systems (e.g., apparatuses) and methods for growing vegetation and/or support aquatic organisms, the system includes a pole assembly comprising one or more hydroponic assembly coupling locations, an aquaculture assembly comprising a water supply operatively coupled to the pole assembly; and one or more hydroponic assemblies. Each of the one or more hydroponic assemblies includes a grow tray. The one or more hydroponic assemblies are operatively coupled to the pole assembly at the one or more hydroponic assembly coupling locations. The grow tray includes a perimeter wall and receives water from the pole assembly adjacent a first portion of the perimeter wall, transfers the water to adjacent a second portion of the perimeter wall, and thereafter transfers the water back to the pole assembly. In some embodiments, the grow tray includes an inner wall, and outer wall, and side walls, and the grow tray receives water from the pole assembly, transfers the water to adjacent the outer wall, and

thereafter the water is transferred back to the pole assembly. The flow of water through the tray allows the water to travel over seeds or roots of vegetation.

[0006] According to some embodiments, a self-contained hydroponics system includes: a water tank; a pump; a supply manifold; a drain manifold; and at least one hydroponics system tower, where the at least one hydroponics system tower includes a pole defining a plurality of pole cups, each pole cup having attached thereto two or more trays, where water is provided from the water tank at a top of the at least one hydroponics system tower by the supply manifold, where the water descends down the at least one hydroponic system tower, entering each of the two or more trays at each pole cup and returning to the at least one hydroponics system tower, where the water exits the at least one hydroponics system tower into the drain manifold.

[0007] According to some embodiments, the at least one hydroponics system tower is supported by a support bracket having a support bracket cup, where water exits a bottommost pole cup into the support bracket cup, wherein the support bracket cup drains to the drain manifold. According to some embodiments, the at least one hydroponics system tower is held at a top of the at least one hydroponics system tower by an upper support bracket having an upper support bracket cup, where the water is provided by the supply manifold to the upper support bracket cup and into the at least one hydroponics system tower.

[0008] According to certain embodiments, the support bracket is secured to a structural frame member to suspend the at least one hydroponic system tower above a floor. According to some embodiments, the lower support bracket provides cantilevered support to the at least one hydroponic system tower. According to certain embodiments, the upper support bracket is secured to the structural frame member to hold the at least one hydroponic system tower between the upper support bracket cup and the lower support bracket cup. The drain manifold, in some embodiments, returns the water to the water tank. The two or more trays at each pole cup extend, in some embodiments, around more than half of a circumference of each pole cup.

[0009] Embodiments provided herein include a hydroponics system including: a water tank; a pump; a supply manifold; a drain manifold; a plurality of lower support brackets, each lower support bracket having a lower support bracket cup; a plurality of upper support brackets, each upper support bracket having an upper support bracket cup, where water is supplied from the water tank to each upper support bracket cup via the supply manifold, where water is returned to the water tank from the lower support bracket cup via the drain manifold. The self-contained hydroponics system of an example embodiment further includes a hydroponics system tower disposed between a respective pair of an upper support bracket cup and a lower support bracket cup, where water flows from the upper support bracket cup, through the hydroponics system tower, and out of the lower support bracket cup. According to some embodiments, the plurality of lower support brackets are cantilevered from a structural frame member. According to some embodiments, the hydroponics system tower defines a central axis, where hydroponic grow trays are disposed about the hydroponics system tower encircling the central axis.

[0010] Embodiments provided herein include a tray for a hydroponics system including: a ramp for guiding water from a water source into a planting area of the tray; at least one channel for returning water from the planting area of the tray to a spout exiting the tray; and at least one water flow control mechanism within the tray, where the at least one water flow control mechanism determines, in combination with a flow rate of water from the water source, a pooling depth of the water within the planting area of the tray. According to some embodiments, the at least one flow control mechanism includes a gate, where the gate is adjustably positioned between the planting area and the at least one channel. According to certain embodiments, the gate, in a first position, allows water to flow from the planting area, under the gate, and into the at least one channel, where the gate in the first position causes pooling in the planting area. According to certain embodiments, the gate, in a second position, allows water to flow from the planting area over the gate, and into the at least one channel, where the gate in the second position causes pooling in the planting area at

least to a height of a top of the gate relative to a bottom of the planting area.

[0011] The tray for a hydroponics system of some embodiments further includes a plurality of protrusions positioned between the planting area and the at least one channel, where the plurality of protrusions permit water flow between them. According to some embodiments, the tray further includes an entry channel where water entering the tray is guided from the ramp to the entry channel before entering the planting area of the tray. The tray of some embodiments further includes a wall separating the entry channel from the planting area of the tray, where the wall is at least partially permeable by water to permit flow from the entry channel into the planting area. The wall of an example embodiment includes a sawtooth shape, where water flows between teeth of the sawtooth shape from the entry channel into the planting area. The water flow of an example embodiment flows between the teeth of the sawtooth shape is substantially similar along a length of the wall.

[0012] The tray of an example embodiment includes a substantially rectangular shape, where the entry channel is disposed along a first side of the tray, water flows across the planting area to a second side of the tray opposite the first side of the tray, and water returns along the channel for returning water from the planting area along a third side of the tray, where the ramp is disposed between the first side of the tray and the third side of the tray. According to some embodiments, the tray further comprises a lid, where the lid defines at least one aperture for receiving therein a pod. According to certain embodiments, a pod supported in the at least one aperture, is configured to wick water from the planting area into the pod. The tray of an example embodiment includes a lid defining a plurality of apertures, each configured to receive therein a pod. The tray of an example embodiment further includes a plurality of flow directing fingers, where the flow directing fingers receive water from the water source and direct the water to pods received in each of the plurality of apertures

[0013] The features, functions, and advantages that have been discussed can be achieved independently in various embodiments or may be combined in yet other embodiments; further details of which can be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0014] Having thus described certain embodiments of the present disclosure in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0015] FIG. 1 depicts a perspective illustration of a hydroponics system in accordance with an example embodiment of the present disclosure;

[0016] FIG. 2 depicts a perspective view of a tank as illustrated in FIG. 1 according to an example embodiment of the present disclosure;

[0017] FIG. 3 depicts a perspective view of a tank as illustrated in FIG. 1 according to an example embodiment of the present disclosure;

[0018] FIG. 4 depicts a perspective view of a strut as illustrated in FIG. 1 according to an example embodiment of the present disclosure;

[0019] FIG. 5 depicts a perspective view of a tank cover as illustrated in FIG. 1 according to an example embodiment of the present disclosure;

[0020] FIG. 6 depicts a perspective view of an aquaculture assembly as illustrated in FIG. 1 according to an example embodiment of the present disclosure;

[0021] FIG. 7 depicts a side view of the aquaculture assembly of FIG. 6 according to an example embodiment of the present disclosure;

[0022] FIG. 8 depicts a perspective view of an aquaculture assembly according to an example

embodiment of the present disclosure;

[0023] FIG. **9** depicts a perspective view of a pole as illustrated in FIG. **1** according to an example embodiment of the present disclosure;

[0024] FIG. **10** depicts a perspective view of the bottom of a pole as illustrated in FIG. **9** according to an example embodiment of the present disclosure.

[0025] FIG. **11** depicts a perspective view of the top of a pole as illustrated in FIG. **9** in accordance with an example embodiment of the present disclosure;

[0026] FIG. **12** depicts a side view of a pole cap as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0027] FIG. **13A** illustrates a perspective view of a pole assembly and a hydroponic assembly as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0028] FIG. **13B** illustrates a perspective view of a pole assembly and a hydroponic assembly as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0029] FIG. **13C** illustrates a perspective view of a pole assembly and a hydroponic assembly as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0030] FIG. **13D** illustrates a perspective view of a pole assembly and a hydroponic assembly as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0031] FIG. **13E** illustrates a perspective view of a pole assembly and a hydroponic assembly as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0032] FIG. **14** illustrates a partial cross-sectional side view of a pole assembly as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0033] FIG. **15** depicts a top side view of a tray as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0034] FIG. **16** depicts a bottom perspective side view of a tray as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0035] FIG. **17A** illustrates a perspective view of a tray as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0036] FIG. **17B** illustrates a pole assembly and a hydroponic assembly in accordance with an example embodiment of the present disclosure;

[0037] FIG. **18A** illustrates a perspective view of a tray as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0038] FIG. **18B** illustrates a pole assembly and a hydroponic assembly in accordance with an example embodiment of the present disclosure;

[0039] FIG. **19** illustrates a pole assembly and a hydroponic assembly in accordance with an example embodiment of the present disclosure;

[0040] FIG. **20** illustrates a pole assembly and a hydroponic assembly in accordance with an example embodiment of the present disclosure;

[0041] FIG. **21** illustrates a pole assembly and a hydroponic assembly in accordance with an example embodiment of the present disclosure;

[0042] FIG. **22** illustrates a hydroponic assembly in accordance with an example embodiment of the present disclosure;

[0043] FIG. **23** illustrates a hydroponic assembly in accordance with an example embodiment of the present disclosure;

[0044] FIG. **24** illustrates a hydroponic assembly in accordance with an example embodiment of the present disclosure;

[0045] FIG. **25** illustrates a perspective view of a tray as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0046] FIG. **26** illustrates a perspective view of a tray member as illustrated in FIG. **25** in accordance with an example embodiment of the present disclosure;

[0047] FIG. **27** illustrates a perspective view of a tray member as illustrated in FIG. **25** in

accordance with an example embodiment of the present disclosure;

[0048] FIG. **28** illustrates a front side perspective view of a spout as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0049] FIG. **29** illustrates a rear side perspective view of a spout as illustrated in FIG. **1** in accordance with an example embodiment of the present disclosure;

[0050] FIG. **30** illustrates a pole assembly and a hydroponic assembly in accordance with an example embodiment of the present disclosure;

[0051] FIG. **31** illustrates a tray and the flow paths according to an example embodiment of the present disclosure;

[0052] FIG. **32** illustrates the tray of FIG. **31** with the return flow paths according to an example embodiment of the present disclosure;

[0053] FIG. **33** illustrates the water supply and return paths from a pole section according to an example embodiment of the present disclosure;

[0054] FIG. **34** illustrates peripheral channels and a flow control mechanism according to an example embodiment of the present disclosure;

[0055] FIG. **35** illustrates the flow control mechanism of FIG. **34** with a flow control barrier according to an example embodiment of the present disclosure;

[0056] FIG. **36** illustrates another tray and the flow paths through the tray according to an example embodiment of the present disclosure;

[0057] FIG. **37** illustrates a tray lid for a tray for receiving pods therein according to an example embodiment of the present disclosure;

[0058] FIG. **38** illustrates a section view schematic of a single pod aperture within the tray lid including a pod according to an example embodiment of the present disclosure;

[0059] FIG. **39** illustrates a tray including flow directing fingers configured to guide water from the pole section to the individual pod locations according to an example embodiment of the present disclosure;

[0060] FIG. **40** illustrates a tray including a tray lid and flow directing figures in profile view according to an example embodiment of the present disclosure;

[0061] FIG. **41** illustrates a trough for distribution of water within a tray according to an example embodiment of the present disclosure;

[0062] FIG. **42** illustrates a detail view of the trough within a tray according to an example embodiment of the present disclosure;

[0063] FIG. **43** illustrates a tray insert that is configured to facilitate water flow within a tray according to an example embodiment of the present disclosure;

[0064] FIG. **44** illustrates a pole cup according to an example embodiment of the present disclosure;

[0065] FIG. **45** illustrates a pair of pole cups interlocked and a flow path there between according to an example embodiment of the present disclosure;

[0066] FIG. **46** illustrates an engagement mechanism between a pole cup and a tray during attachment according to an example embodiment of the present disclosure;

[0067] FIG. **47** illustrates an engagement mechanism between a pole cup and a tray after attachment according to an example embodiment of the present disclosure;

[0068] FIG. **48** illustrates a tray secured to a pole cup according to an example embodiment of the present disclosure;

[0069] FIG. **49** illustrates five trays arranged around a single pole cup according to an example embodiment of the present disclosure;

[0070] FIG. **50** illustrates a top view of four substantially square trays arranged around a pole cup according to an example embodiment of the present disclosure;

[0071] FIG. **51** illustrates a perspective view of the square tray arrangement of FIG. **50** according to an example embodiment of the present disclosure;

[0072] FIG. **52** illustrates a mobile tower of pole cups and trays for a hydroponic system according to an example embodiment of the present disclosure;

[0073] FIG. **53** illustrates an example embodiment of a hydroponic system within a confined space such as a container according to an example embodiment of the present disclosure;

[0074] FIG. **54** illustrates a base of a tower of pole cups and trays of a hydroponic system according to an example embodiment of the present disclosure;

[0075] FIG. **55** illustrates a top of a tower of pole cups and trays of a hydroponic system according to an example embodiment of the present disclosure;

[0076] FIG. **56** illustrates a pump and tank arrangement for a hydroponic system described herein according to an example embodiment of the present disclosure; and

[0077] FIG. **57** illustrates a containerized embodiment of a hydroponic system according to an example embodiment of the present disclosure.

DETAILED DESCRIPTION

[0078] The present disclosure now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments are shown. Indeed, this disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout.

[0079] Embodiments described herein can be used for both hydroponic and aquaponic environments. The systems of example embodiments will generally be described as hydroponic; however, it is understood that a hydroponic system can be adapted to be an aquaponic system. As such, the use of the term hydroponic is not to be limited to exclusively hydroponic systems, but hydroponic systems that can be adapted to aquaponic systems as described further below.

[0080] FIGS. **1-59** illustrate embodiments of the hydroponic system **1**, and the various components thereof. As illustrated in FIG. **1**, the hydroponic system **1** includes, in general, three assemblies: 1) a tank **101** which in an aquaponic system can be an aquaculture assembly and 2) one or more hydroponic assemblies **200** operatively coupled with the tank **101** via 3) a pole assembly **300**. The various assemblies, and components thereof, of the hydroponic system **1** of FIG. **1** are illustrated in FIGS. **2-31** and described in further detail throughout this specification. In general, the tank **101** is located at the base of the hydroponic system **1** and in an aquaponic system is configured for housing aquatic creatures in a controlled marine environment. The tank **101** of example embodiments can be co-located with the pole assembly **300**, such as in the embodiment of FIG. **1**; however, a tank can be remotely located from the pole assembly **300** and hydroponic assemblies **200** as described below with respect to various embodiments. Regardless of the location of the tank **101**, the system can be employed as a hydroponic system or aquaponic system.

[0081] The hydroponic system **1** of the illustrated embodiment of FIG. **1** includes a tank **101** having a base plate **107** positioned at the bottom of the tank **101**, and one or more tank covers **114** positioned at the opening of the tank **101**. The pole assembly **300** is at least partially submerged in the tank **101** and extends generally vertically from the base plate **107** of the tank **101** in an upright orientation. The pole assembly **300** (or additional components within) is configured for supplying water from the tank **101** to the hydroponic assemblies **200**, and is further configured to control the flow of water as it dispenses to and from the hydroponic assemblies **200**, and thereafter returns to the tank **101**.

[0082] The pole assembly **300** may of the illustrated embodiment includes a pole **301**, a plurality of pole cups **305** located vertically along the pole **301**, and a pole cap **312** or one or more openings **304** positioned at the top of the pole **301**. The hydroponic assemblies **200** may be operatively coupled to the pole assembly **300** and configured for housing plants in a controlled growing environment wherein the hydroponic assemblies **200** may receive water from the pole assembly **300** via a plurality of spouts **400** or various other means described in further detail below. The pole assembly may further include one or more hydroponic assembly locations located vertically about

the pole and configured to receive the hydroponic assemblies **200**. In a preferred embodiment, the pole cups **305** function as hydroponic assembly coupling locations; however, in other embodiments, the pole **301** itself may be a hydroponic assembly coupling location such that the hydroponic assemblies **200** may be operatively coupled with the pole assembly **300** at any location and in any orientation about the vertical axis of the pole **301** using various couplings mechanisms (e.g., slotted holes through the pole and a tab or hook on the tray, an apron or lip on the pole and a tab or hook on the tray, slots or grooves on the surface of the pole and a projection on that tray that slides within the slots or grooves, or any other type of coupling).

[0083] The hydroponic assemblies **200** may include one or more trays **201** where the trays **201** optionally have a tray **215** operatively coupled to the trays **201**, which may be deposited within the trays **201**, located on top of the trays **201**, span an opening in the trays **201**, or otherwise be attached to the trays **201**. In some embodiments, the hydroponic system **1** may include an optional tube (not illustrated) located within the pole **301** to supply water from the tank **101**, throughout the pole **301**, and to the hydroponic assemblies **200**.

[0084] With respect to the embodiment of FIG. **1**, and as discussed in further detail throughout this specification water is pumped up through the pole **301** (e.g., a tube within the pole **301** or through the pole **301** itself) and exits the pole cap **312** or one or more openings **304** at or near the top of the pole **301**. The water flows down the outside surface of the pole **301** and is diverted by the spouts **400**, which direct the water from the outside surface of the pole **301** to the trays **201** and/or trays **215**. As will be discussed in further detail later the water will either flow into the tray **215** and then into the back of the tray **201**, or flow directly into the back of the tray **201** in order to provide a continuous stream of water to the plants in the trays **201** and/or trays **215**. The water on the bottom of the tray **201** then flows back to the pole **301** and/or cup **305** and is collected in a second cup **305** located vertically below the first cup **305**. The cup **305** funnels the water back to the outside surface of the pole **301** for delivery to a second tray **201** and/or tray **215** located below the first tray **201** and/or tray **215**. The water then flows back to the tank **101** and the process begins again.

[0085] FIGS. **2** through **8** illustrate the tank **101** and its components in accordance with an example embodiment of the present disclosure. As illustrated in FIGS. **2** through **8** the tank **101** includes a base plate **107** adjacent to a bottom of the tank **101**, and one or more tank covers **114** adjacent to the opening **103** of the tank **101**, wherein the tank covers **114** may be positioned at or above an outer perimeter of the tank **101** using a plurality of struts **120** as depicted in FIG. **4**.

[0086] As illustrated in FIGS. **2** through **3**, the tank **101** may be defined by a cylindrical reservoir that has a base **102**, an opening **103**, and a perimeter wall **104**. The outer perimeter of the base **102** may be adjacent to bottom edge of the perimeter wall **104** such that the opening **103** is defined by a recess formed therein. The opening **103** may further have an outermost perimeter **105** that is defined by the top edge of the perimeter wall **104** of the tank **101**. The tank **101** may additionally comprise one or more ridges **106** that extend outwardly from the perimeter **105** of the tank opening **103** such that the ridges **106** are configured for receiving at least a portion of the struts **120** to stabilize the pole **301** within the tank **101**. For example, as shown in FIG. **2** and FIG. **3**, the tank **101** may include three (3) ridges **106** that are equally distanced from one another, along the perimeter **105** of the tank opening **103**, and slightly extend outward beyond the perimeter wall of the tank **101c**. As illustrated in FIG. **1** and FIG. **7**, the ridges **106** are configured to receive at least three (3) struts **120**, where one (1) strut **120** is operatively coupled with one (1) corresponding ridge **106**.

[0087] It should be noted that although in some embodiments the tank **101** is defined by a cylindrical shape, the tank **101** may be embodied by other shapes that are illustrated and are not illustrated herein, including but not limited to, cubed, rectangular, trapezoidal, and/or organic or ameba shapes. In said embodiments, the tank **101** may have a plurality of perimeter walls as opposed to the one continuous perimeter wall **104** shown in the illustrated embodiments. For example, as shown in FIG. **8**, the tank **101** may be rectangular in shape such that it includes four

(4) perimeter walls and four (4) corresponding tank ridges configured to receive struts **120** and stabilize the pole assembly **300** within the tank **101**. It should be understood that any size and shape may be utilized for the tank **101** within the hydroponics system **1**. As further illustrated in FIG. **8**, in some embodiments, the struts **120** may be adjustable to accommodate a variety of tank **101** shapes. In such an embodiment, the struts **120** may be embodied by two individual members that can be coupled with each other using a fastener **120a** and/or another coupling mechanism. In this way the struts **120** may be retractable and/or extendable based on the size of the tank. In some embodiments, the system may use a combination of static (one size) and dynamic (retractable and/or extendable) struts **120**.

[0088] As further illustrated in FIGS. **2** through **8**, the base plate **107** may be positioned adjacent to the base **102** of the tank **101** such that the base plate **107** rests above the base **102** of the tank **101**. The base plate **107** may be defined by a circular shaped plate that includes a pole anchor **108** that extends upwardly from the center of the base plate **107**. In other embodiments, the base plate **107** may be of a thickness that allows the pole anchor **108** to extend inwardly into the base plate **107**. In one embodiment, the base plate **107** is a plate that is removably coupled with the base **102** of the tank, and in another embodiment, the base plate **107** is integral within the base **102** of the tank **101** itself. The pole anchor **108** may be operatively coupled with the base plate **107** (e.g., glued, locked into place, formed integrally within, or otherwise secured to the base plate **107**). As shown in FIG. **2** and FIG. **3**, the base plate **107** may reflect the shape of the tank **101** such that in other embodiments the base plate **107** may be embodied by other shapes not illustrated herein, including but not limited to, cubed, rectangular, trapezoidal, and/or organic or ameba shapes. Additionally, in other embodiments, the base plate **107** may be defined by a shape that is different from the shape of the tank **101**. For example, a rectangular tank **101** may comprise a circular base plate **107** therein.

[0089] The pole anchor **108** within the base plate **107** may be defined by a recess **109**, an access notch **110**, an inner wall **111**, an outer wall **112**, and/or a top edge **113**. The recess **109** may be centered in the middle of the pole anchor **108** and configured for receiving the bottom of a pole **301** such that the pole **301** is stabilized in an upright position after being placed into the recess **109**. To this extent, the shape and size of the recess **109** may be defined by the inner wall **111** of the pole anchor **108** such that it reflects the shape of the pole **301**. Additionally, the pole anchor **108** may have an access notch **110** that extends from the inner wall **111** of the pole anchor **108** to the outer wall of the pole anchor **108**. The access notch **110** may allow for an optional tube to pass through the pole anchor **108** and be redirected into the pole **301** permitting the tube to supply water from the aquaculture assembly, throughout the pole assembly **31** and, to the hydroponic assemblies **200**. In some embodiments, as illustrated at least in FIGS. **13a-13b** and FIG. **30**, the tube may be connected to a pumping mechanism that pumps water, from the aquaculture assembly, into the tube. As such in an alternative embodiment, the access notch **110** may be omitted and the base plate **107** may further comprise a hole in the bottom of the base plate **107** that aligns with the hole of an opening in the bottom of the pole **301** to permit the supply water into the pole assembly **300**.

[0090] As illustrated in FIG. **4**, the struts **120** may be several elongate structures that are configured for bracing the pole **301** after it has been operatively coupled with the base **102** of the tank **101** (e.g., through the pole anchor **108**). As shown in the illustrated embodiments, the struts **120** may be at least partially defined by a first end **121** and a second end **122** that are operatively coupled (e.g., configured to connect and/or fasten the struts **120**) to the pole assembly **300** and the tank **101**. In one specific embodiment, the first end **121** of the strut **120** may fasten to at least a portion of the pole cup **305**, and the second end **122** may fasten to a ridge **106** of the tank **101** such that the pole **301** is stabilized in an upright position and less likely to be deflected out of the generally vertically orientation. As such the height of the pole cup **305** with respect to the ground (e.g., when installed into the pole anchor **108**) may be equivalent to the height of the tank **101**. Furthermore, the first and second ends of the struts (**120a**, **120b**, respectively) may comprise grooves and/or other coupling mechanisms that allow the ends to fasten and conform to either a hydroponic assembly

the pole cups **305**, or a ridge **106** of the tank **101**. In some embodiments, the struts **120** may be positioned such that they are equally distanced from one another and exert an equivalent force on both the pole **301** and the tank **101**.

[0091] The struts **120** may be further defined by at least one or more indentations **123** (e.g., rails, shelves, cover supports, or the like), in the top surface of the strut **120**, for receiving the tank cover **114**. As shown in FIG. 4, in one embodiment where the tank **101** has multiple tank covers **114**, the strut **120** may comprise two (2) indentations **123**. FIG. 1 further illustrates where, in such an embodiment, at least a portion of a first tank cover **114** may be positioned adjacent to the first indentation **123** and at least a portion of a second tank cover **114** may be positioned adjacent to the second indentation **123** such that the top surfaces of the tank covers **114** are flush with the top surface of the strut **120**. In another embodiment where the tank **101** has a single tank cover **114**, the strut **120** may comprise only one (1) indentation **123** configured to receive at least a portion of the tank cover **114** therein. In one embodiment, the length of the strut **120** may be equivalent to the distance from the ridge **106** of the tank **101** to an edge of the cup **305** on the pole **301**. In one embodiment, as illustrated in FIG. 1, where the tank **101** is cylindrical in shape, the length of the strut **120** may be less than the radius of the tank **101**. In another embodiment, as illustrated in FIG. 8, where the tank **101** is rectangular in shape, the length of the struts **120** may vary with respect to the distance between the tank wall and the edge of the cup **305** of the pole **301**. For example, the length of two of the struts **120** (out of the four struts **120** illustrated in FIG. 8) operatively coupled to the rectangular tank **101** with respect to its width may be sustainably less than the length of two of the struts **120** applied to the tank with respect to its length. It should be understood that any type of strut **120** may be utilized in order to help support the pole **301** within the tank **101**.

[0092] As illustrated in FIG. 5, the one or more tank covers **114** may be embodied by flat plates that are configured to be positioned adjacent to the opening **103** of the tank **101** using one or more struts **120** such that the top surface of the tank covers **114** are flush with the top surface of the struts **120** and/or the opening perimeter **105** of the tank **101**. As shown in FIG. 5, in one embodiment, the tank covers **114** may comprise several notches **115** on the corners of each of the individual tank covers **114** that allow the covers **114** to be received in the indentations **123** within the top surface of the struts **120**. In this configuration, the inner edges of the tank covers **117** are adjacent to the pole **301**, the outer edges of the tank covers **118** are adjacent to the opening perimeter **105** of the tank **101**, and the side edges **119** are positioned adjacent to barriers that separate the indentations **123** in the top surface of the struts **120**. In the illustrated embodiments, the area **116** of the tank cover **114** that extends from the notches **115** is equivalent to the area of the indentation **123** within the top surface of the struts **120** such that when positioned adjacent to the top of the strut the tank covers **114** fit securely in place. In an embodiment, where the tank **101** is cylindrical in shape and includes several tank covers **114**, the tank covers **114** may be arc shaped such that when positioned at the opening **103** of the tank **101** the covers **114** conceal a majority portion of the opening **103** of the tank **101**. In one embodiment, the tank **101** may comprise a single tank cover **114** such that when positioned adjacent to the opening **103** of the tank **101** using the one or more struts **120** the top surface of the tank covers **114** rest above the top surface of the struts **120**. The tank covers **114** may additionally comprise notches located along the outer boundary of the tank cover **114** and adjacent to the opening perimeter of the tank **101d** that allow for visibility into the tank **101** as well as a means to easily remove and place the tank cover **114** on top of the struts **120**, feed any organisms living the tank, supply nutrients to the water, or the like.

[0093] FIGS. 9 through 14 illustrate a pole assembly **300** in accordance with one or more embodiments of the present disclosure. As illustrated in FIGS. 9 through 14 the pole assembly **300** may comprise a pole **301** having one or more pole cups **305** arranged about the pole **301** in a vertical orientation, and a pole cap **312** or opening **304** located at the top of the pole **301**, wherein the pole cap **312** is at least partially coupled with the pole **301**.

[0094] As illustrated in FIGS. 9 through 14, the pole **301** may be defined by an elongate conduit

that has a pole extension **302** at the top of the pole **301**, an access notch **303** located at the bottom of the pole, and an opening **304** located at the top of the pole. The access notch **303** may be provided for allowing the passage of an optional tube into the pole **301** such that the tube may supply water from the aquaculture assembly, throughout the pole **301** and, to the hydroponic assemblies **200**. In an embodiment, as illustrated in FIG. 7, the bottom of the pole **301** is positioned in the pole anchor **108** of the base plate **107** such that the base plate access notch **110** and the pole access notch **303** are aligned with one another and form an opening that extends from the tank opening **103** to the interior of the pole **301**.

[0095] The pole extension **302** may be slightly narrower than the pole **301** such that it can be coupled with a pole cap **312**. In this regard, as illustrated in FIG. 1, the pole cap **312** may be configured to receive the pole extension **302** in an opening of the pole cap **312**. The pole cap **312** may have a plurality of exit slots **313** located at the top of the pole cap **312** that allow the water to exit the interior of the pole and be distributed to the outside surface of the pole **301**, the spouts **400**, the hydroponics assemblies **200**, back to the pole cups **305**, and back down outside surface of the pole **301**. To this extent the size of the openings of the exit slots **313** may allow the water to flow at a controlled rate as it exits the pole **301**. The pole cap **312** also prevents water from shooting straight into the air, and may also have channel surfaces **315** to transition the water to the outside surface of the pole **301**.

[0096] In an alternative embodiment, as shown in FIG. 13A, the exit slots **317** may be located in the pole **301** itself versus being in the pole cap **312**. In one embodiment, the exit slots **317** may be a fixed size. However, in other embodiments the size of the exit slots **317** may be adjusted to smaller increments, and/or the exit slots **317** may be manually or automatically opened or closed such the flow of the water may be regulated with a slider located either inside or outside the pole **301** or pole cap **312**. The pole cap **312** may be provided to increase or decrease the pressure within the pole **301** and/or the tube that may run inside of the pole **301**. As such, the pole cap **312** may be adjusted to increase or decrease the flow of water that is allowed to exit the pole **301** (or tube within the pole **301**).

[0097] In one embodiment, the pole cap **312** may additionally have an opening **314** for receiving a fastener (e.g. a screw, or the like) to secure the pole cap **312** to the pole **301** and/or the tube. In this regard, both the pole cap **312** and the pole **301** and/or the tube may have an opening **314** that is internally threaded such that the openings **314** can be aligned with one another and receive a fastener. In another embodiment, the pole cap **312** may screw directly onto the pole **301**. To this extent, at least a portion of the pole cap **312** may be internally threaded, and at least a portion of the pole extension **302** may be externally threaded such that the pole extension **302** is configured to receive the pole cap **312**. In yet another embodiment, the pole cap **312** snaps, slides, clips, or the like onto the pole extension **302** such that it is secured in place and no additional threading is necessary. In an alternative embodiment, the pole cap **312** may be omitted from the design and the pole assembly **300** may use an alternative means for building pressure within the pole **301** and controlling the flow of water out of the pole **301**. For example, as shown in FIG. 13, the top of the pole **301** may be enclosed versus having an opening, and the pole may comprise several exit slots **317** to allow water to exit the pole **301**.

[0098] As further illustrated in at least FIGS. 9 through 11, the pole **301** may comprise one or more pole cups **305** that are arranged along the pole **301** with respect to its vertical axis. The one or more pole cups **305** may be divided into upper and lower tiers (**306**, **307**, respectively). The upper tier **306** of the pole cup **305** may be configured for receiving an overflow of water as it is being dispersed from another area within the system (e.g., a pole cup **305** located above) and funneling the water to the lower tier **307** of the pole cup **305b**. As shown in FIGS. 9 through 11, in one embodiment, the lower tier **307** may be narrower than the upper tier **306**. It should be noted that although in the illustrated embodiments, the pole cups **305** are generally circular in shape, the pole may be embodied by other shapes not illustrated herein, including but not limited to, cubed,

rectangular, trapezoidal, and/or organic or ameba shapes. The pole assembly **300** may additionally comprise a plurality of blades **311** (e.g., also described as fans, diffusers, diverters, or like) that control the rate at which the water exits the lower tier of the cup and flows down the pole to the hydroponic assembly **200**, another cup **305**, or to the tank **101**. In a preferred embodiment, as illustrated in FIG. **10**, the blades **311** may be located between the outer perimeter of the pole **301** and the inner perimeter of the lower tier **307** of the cup **305**. The blades **311** may extend outward from the outer perimeter of the pole **301** until they reach the inner perimeter of the lower tier **307** of the cup **305** and upward such that the blades **311** are angled and/or sloped with respect to the ground level. The blades **311** may further be angled towards the direction of the outer surface of the pole **301** to help direct the water onto the surface of the pole **301**. It should be noted that the blades **311** may be positioned in other configurations not shown in the illustrated embodiments such that the configuration of the blades **311** may vary the rate of flow of the water (e.g., slow down or speed up the flow of water).

[0099] In one embodiment, the one or more pole cups **305** may be permanently coupled to the pole **301**, for example formed integrally with the pole **301** (e.g. molded, plastic welded, or the like) or permanently secured to the pole **301**. The pole cups **305** may be located at the desired height in order to space apart the trays **201** to grow different types of plants or other vegetation (e.g., vegetables, herbs, spices, fruits, grasses, flowers, plants, or any other type of editable or non-editable vegetation). In another embodiment, as shown in FIG. **14**, the one or more pole cups **305** may be removably coupled to the pole **301**. In an embodiment, where the pole cups **305** are removable the pole cups **305** may further comprise a hook and/or clasp mechanism to connect the cup **305** to the pole **301**. In one example, a half section of the cup is removable by detaching a hook **308** on a first end of the pole cup **305** from the attachment point **309** located on the second and/or opposite end of an adjacent pole cup **305**. It should be that, although a hook and/or clasp mechanism is shown in the illustrated embodiment, other methods may be used to secure to two halves of the pole cup **305** and pole **301** together including other fasteners (e.g. screws). The pole cup **305** may further comprise inner bracing **310** to stabilize the pole cup **305** as it is attached to the pole **301**. Additionally, the pole **301** may be asymmetrically configured such that it is defined by a series of curvatures where the “in and out” orientation of the curves prevents the pole cup **305** from easily sliding down the pole **301**. It should be noted that although in the illustrated embodiments, the pole is generally cylindrical in shape, the pole may be embodied by other shapes not illustrated herein, including but not limited to, cubed, rectangular, trapezoidal, and/or organic or ameba shapes.

[0100] In other embodiments of the invention the pole cup **305** may slide over the pole **301**, and a section of the pole cup **305** may be tightened, secured to with a faster, or otherwise be attached to the pole **301**. In still other embodiments, the diameter of the pole **301** may increase gradually or may otherwise have different diameters along the pole **301**, such that pole cuts with different diameter holes may be placed over the top of the pole **301** and located on the pole **301** based on the diameter of the pole **301** and the inner diameter of the pole cup **305**.

[0101] FIGS. **15** through **26** illustrate the hydroponic assembly **200** in accordance with various embodiments of the present disclosure. As illustrated in FIGS. **15** through **26** the hydroponic assembly **200** includes one or more grow trays **201**, wherein the grow trays may optionally have a tray **215** placed within the tray **201**.

[0102] As illustrated in FIGS. **15** through **27**, the grow tray **201** may be defined by a tray that has a plurality of protrusions **202** extending upward from the base of the tray **201**, at least one channel **205** having a channel opening **206**, at least one sloped surface **207** in the base of the tray **201b**, one or more hooks **208**, and side rails **209a** and **209b**. As shown in the figures, in one embodiment, the tray **201** may be further defined by walls that extend upward from the base of the tray **201**. The tray **201** may comprise an inner wall **212**, an outer wall **213**, and two side walls **214a**, **214b** that collectively form an opening in the top of the tray.

[0103] In one embodiment, the side rails **209a**, **209b** may be located parallel to the interior of the side walls **214a**, **214b** such that two channels **205a**, **205b** are formed. In the illustrated embodiments, a first channel **205a** is formed on the left side of the tray **201**, and a second channel **205b** is formed on the right side of the tray. However, in other embodiments, the tray may comprise only one side rail **209a** such that only one channel **205a** is formed. The side rails **209a**, **209b** may comprise an inner edge **210** and an outer edge **211**, where the inner edge **210** is adjacent to the inner wall **212** of the tray **201**, and the side rails **209a**, **209b** may extend from the inner wall **212** of the tray **201** until the outer edge **211** of the side rail **209** is adjacent (e.g., within a short distance of) the outer wall **213** of the tray **201** such that a gap is formed between the outer edge **211** of the side rail **209** and the outer wall **213** of the tray **201**. Each channel **205a**, **205b** may lead to a channel opening **206a**, **206b** that is formed in the inner wall **212** of the tray **201** that opens to external surface of the pole cup **305** or the external surface of the pole **301**.

[0104] As such, in one embodiment of the invention, as water flows into the base of the tray **201** the water is first directed to the outer wall **213** because the base is sloped towards the outer wall, flows through the gaps, and back down the one or more channels **205a**, **205b** towards the pole **301** because the channels **205a**, **205b** are sloped towards the pole **301**. As such, the tray **201** provides a means for water to continuously enter the base of the tray **201**, flow over the seeds or roots of the plants or other vegetation towards the outer wall **213**, and back towards the inner wall **212** adjacent the pole **301** through the channels **205a**, **205b**.

[0105] In one embodiment, upon exiting the tray **201** the water may flow towards the exterior of the lower tier **307** of a pole cup **305** and travel down the exterior of the pole cup **305** and subsequently continue to flow down the pole **301**. In another embodiment, the lower tier **307** of a pole cup **305** may comprise an opening adjacent to the channel opening such that the water flows directly into the interior of the lower tier **307** of the pole cup **305** and through the plurality of blades **311** before exiting the pole cup **305**. The tray **201** may additionally comprise at least one sloped surface **207** in the base **203** of the tray **201** that defines the direction in which the water should flow. For example, as shown in the illustrated embodiments of FIGS. **15-18**, the tray may comprise two sloped surfaces (**207a**, **207b**) that extend into the base **203** of the tray at an angled depth such that it creates an incline in the channels **205a**, **205b**, where the channels **205a**, **205b** slope downward from the outer wall **213** towards the inner wall **212** of the tray allowing water easily to flow towards the pole **301**. Likewise, in some embodiments, a similar inclination may be created in base **203** of the tray **201** such that the base slopes downward from the inner wall **212** towards the outer wall **213** of the tray allowing water to flow towards the outer wall **213** of the tray **201** and subsequently into the at least one channel **205**. In yet another embodiment, the base may be defined by a flat surface that extends from the inner wall **212** of the tray **201** to the outer wall **213** of the tray **201**.

[0106] In another embodiment, as shown in FIG. **22**, two channels (**205a**, **205b**, respectively) may be formed adjacent to the top edges of the side walls (**214a**, **214b**) of the tray on the left and right sides versus the in the base **203** of the tray, wherein the channels **205** may be adjacent to the interior of the side walls **214a**, **214b** (side rails **209a**, **209b**, and protrusions **202** not illustrated). In such an embodiment, the channels **205a**, **205b** may carry the water towards to outer wall **213** of the tray on each side the water may then overflow at the end of the channel **205** and enter the base **203** of the tray, wherein the base **203** of the tray **201** is sloped towards the pole assembly **300** such that the water flows towards the pole assembly **300** and exits the tray **201** through an opening **206** in the inner wall of the tray **201**. In other embodiments there may be other types of channels located within the tray **201** to deliver the water toward a location adjacent the outer wall **213** of the tray **201** and back towards the inner wall **212** of the tray **201**.

[0107] In yet another embodiment, as shown in FIG. **23**, the water may travel throughout the hydroponic assembly **200** and back to the pole assembly **300** without the use of the one or more channels **205** in the tray **201**. As illustrated a first layer in the tray **201** may direct the water flow to

the back of the tray (e.g., towards the outer wall **213**) and a lower layer in the tray **201** may direct the water back to the inner wall **212** of the tray **201**. The first layer in the tray **201** may also have protrusions **202** (not illustrated) as illustrated in FIGS. **15-20**, or the like.

[0108] In another embodiment, as shown in FIG. **24**, one channels **205** may be formed adjacent to the top edges of a side walls of the tray **201** on either the left and right sides versus the in the base **203** of the tray, wherein the channel **205** may be adjacent to the interior of the side walls **214** (side rails **209**, and protrusions **202** not illustrated). In such an embodiment, the channel **205** may carry the water towards the outer wall **213** of the tray such that the water over flows at the end of the channel **205** and at the interior sides of the channel **205** and enters the base **203** of the tray, wherein the base **203** of the tray **201** is sloped downward away from the channel **205** and towards the opposite side wall **214** such that the water flows from the first side wall **214a** to the second side wall **214b** and exits the tray **201** through an opening **206** in the inner wall of the tray **201** that is adjacent to the second side wall **214b**.

[0109] As shown in FIGS. **18a** and **18b**, in one embodiment, the tray **201** may comprise extended drip ramps **206c** that extend from the channel openings **206** in the inner wall of the tray **201**. In such an embodiment, the drip ramps **206c** may extend towards the pole **301** such that water may exit the tray **201** through the openings **206** in the inner wall of the tray **201** and flow directly towards the pole **301** versus flowing towards the exterior of the lower tier of the pole cup, as illustrated in FIG. **18b**. In such an embodiment tray **201** may be positioned such that a gap exist between the drip ramps **206c** and the bottom surface of the pole cup **305** allowing water to freely flow towards the pole **301** without any restriction.

[0110] The plurality of protrusions **202** extending upward from the base of the tray **201** may be located in between the interior of the side rails **209a**, **209b**. As shown in the Figures, in one embodiment, at least a portion of the protrusions **202** may be arranged in equally distanced rows that are parallel to the inner and outer walls (**212**, **213**, respectively) of the tray **201**, where the protrusions **202** are configured such that the roots of a plant are able to at least partially wrap around the protrusions **202** and anchor the plant during the growing process, for example when the hydroponic assembly **200** uses a tray **215**, as will be described in further detail later. In other embodiments of the invention, other vegetation such as wheat grass, or other types of similar vegetation may be grown within the base of the tray **201** between the protrusions **202**. The protrusions **202** may act as locations in which the seeds of various types of vegetation may be placed to provide support for keeping the seeds in place and not washing away with the flow of water. The plurality of protrusions may vary in height and/or width. In one embodiment, the height of any given protrusion **202** does not exceed above the height of the side rails **209a**, **209b**.

[0111] The tray **201** may comprise one or more hooks **208** that allow that tray **201** to be connected to a cup **305** on the pole **301**. It should be noted that the tray **201** may be connected to the pole assembly **300** using various coupling mechanisms or methods. As shown in FIG. **18** through FIG. **20**, in one embodiment, the coupling method is a hooking mechanism that allows the tray **201** to hook onto the upper tier **306** of the pole cup **305**. One or more hooks **208** may extend from the inner wall **212** of the tray **201** and hook onto the top edge of the upper tier **306** of the pole cup such that the inner wall **212** of the tray **201** is adjacent to the exterior of the pole cup **305**. In this regard, the inner wall **212** of the tray **201** may reflect the shape of the pole cup **305** such that it conforms to the pole cup **305** and is flush with the exterior surface of the pole cup **305**. For example, the inner wall **212** may have the same shape as the exterior surface of the upper tier **306** and lower tier **307** of the pole cup **305**. It should be noted that the trays **201** can be moved and spaced in a **360** degree orientation around the pole cup **305** and on different levels of the pole cups **305** to support different types of plants. For example taller tomato plants may have a tray **201** on a lower cup and no tray located above the tomato plant to allow the plant to grow, while lettuce may be grown in a tray **201** that also has lettuce trays above and below. This invention allows the mixing and matching of different configurations to grow any type of vegetation. As such the hydroponics assemblies **200**

may further comprise clips that can clip onto the tray **201**, cup **305**, tray **215** or the like in order to support different types of plants. For example a clip (in some embodiments with a dowel to support the plant) may be used to support a tomato plant to grow vertically.

[0112] Although in the illustrated embodiments, a hooking mechanism is used to couple the tray **201** with the pole cup **305**, other means of connecting the two structures may be used in conjunction with the present invention. In one embodiment, as shown in FIG. **21**, the tray **201** may be connected directly to the pole **301** itself. As such, the pole may comprise several openings and/or exit slots **317** that are configured for receiving the hooks **208** of the tray **201** such that the hooks **208** are at least partially positioned within the interior of the pole **301**. In such an embodiment, the use of a spout **400** may be omitted from the design in order to transport water from the pole **301** to the tray **201**.

[0113] In the alternative, the hooks **208** may contain smooth groove or indentations at the top such that the water flows from the exterior of the pole **301** through the grooves within the hooks **208** and into the base of the tray **201**. In other embodiment the flow of water back down to the tank **101** may occur within the pole **21**, such that water is flowing upwardly through the pole **21** in a first chamber, such as the tube, and running back down the pole **21** in a second chamber, such as between the outside of the tube and the inside of the pole **21**. The hooks **208** may catch the water on the inside of the pole **301** and deliver it to the tray **201**.

[0114] In yet another embodiment, the tray **201** may be connected to the pole **301** via a spout **400**. The spout **400** may be permanently or removably coupled with the pole **301**. As shown in FIG. **13B** and **13C**, in such an embodiment, the spout **400** may be embodied by a pole apron having a plurality of slits therein such that the tray **201** is positioned to rest adjacent to the top edge of the pole cup **305** and wedges underneath a portion of the pole apron such that it is securely positioned in place. The tray **201** may be additionally supported by a cantilever. In other embodiments, the pole apron may have a continuous perimeter wall and exclude the plurality of slits. In some embodiments, the wedge may hook into a portion of the pole **301** of the pole apron (spout **400**). In an exemplary embodiment, at least a portion of the bottom surface of the tray **201** is flush with the top edge of the pole cup **305** to provide a means of stability. The water may then run down the pole **301** and down a top surface of the pole apron (spout **400**), optionally to the plurality of slits and/or through (in between) the slits, and into the tray **201**. As such the tray **201** may comprise an extended inner wall **212** that extends beyond the top surface of the side walls **214**, and the outer wall **213**, and defines a wedge that is configured to wedge beneath at least a portion of the pole apron (spout **400**) and fasten the tray **201** in a secure position. For example the tray **201** may be inserted in the spout **400** at an angle and lock into place using the wedge. In an exemplary embodiment the extended inner wall **212** (and/or wedge) may be $\frac{1}{4}$ inch in length. However, in other embodiment, the length of the extended inner wall **212** and the side walls in general may vary in lengths and/or widths. In some embodiments, the tray **201** may comprise an extended inner wall **212** that extends beyond the side walls **214** outward from the pole in a dovetail configuration, that is configured to horizontally slide into one or more slits within the pole apron. In some embodiments, the system may comprise an additional reverse spout **400** that is connected to either the pole **301** and/or the tray **201** and configured to direct the flow of water from the pole apron back into a pole cup **305** positioned beneath the pole apron.

[0115] As illustrated in FIGS. **25** through **27**, a grow tray **215** may be utilized along with the tray to grow different types of vegetation. The grow tray **215** may be defined by a tray that has a first and second members **216**, **217**, which are configured to be coupled with one another such that a plurality of interior tray channels **218** are formed therein. As shown in the illustrations, in one embodiment, the tray **215** may be further defined by walls that extend upward from the base of the tray **215**. The tray may comprise an inner wall **224**, an outer wall **225**, and two side walls **226a**, **226b** that collectively form an opening in the top of the tray **215**.

[0116] The first and second members **216**, **217**, respectively of the grow tray **215** may

independently have a plurality of interior channels **218**. Each interior channel **218** of the first member **216** may have a channel edge **219** that is configured to be operatively coupled with a corresponding edge hole **221** of the second member **217**. Likewise, each interior channel **218** of the second member **217** may have a channel edge **219** that is configured to fasten into a corresponding fastening hole **221** of the first member **216**. The interior channels **218** may be parallel to one another with respect to the inner and outer walls (**224**, **225**, respectively) of the tray **215** such that when the first and second members (**216**, **217**, respectively) are coupled together a plurality of openings **223** is formed in the base of the tray **215**. The openings **223** may allow water to flow through the base of the tray **215** into a tray **201** positioned below the tray **215**. The tray **215** may additionally have at least one main channel **220**, where each of the main channel **220** may correspond to either the first or second member **216**, **217** of the tray **215** such that when water flows into the tray **215** it may either (a) flow directly through one of the openings in the base of the tray, or (b) flow down the main channel **220** and be dispersed throughout the interior channels **218**. In this regard each interior channel **218** may have an opening, opposite of the channel edge **219**, which is connected to and can receive water from the main channel **220**. Furthermore, the water may overflow out of the interior channels **218** and into the openings **223** between the interior channels **218**. As explained in further detail later the seeds of various plants may be located within the channels **220**, **218**. For example, rocks may be placed in the channels **220**, **218**, and the seeds may be placed in cotton balls between the rocks. The water flows over the seeds and down into the tray, and when the seeds grow the roots will reach down into the tray **201** through the openings **223** in the tray **215**. As further illustrated in FIG. 26, the outer wall **225** of the tray **215** may optionally comprise one or more openings **225a** that allow water to exit from the rear of the tray **215** and enter the tray **201** positioned below. In this way the openings **225a** in the outer wall **225** of the tray **215** may prevent water from remaining stagnant within the tray after it is received from the pole assembly **300** and flows throughout the channels **228** within the tray **215**.

[0117] As illustrated in FIG. 19 and FIG. 20, the shape of the tray **215** after the first and second members **216**, **217** have been coupled with one another may reflect the shape of the tray **201** such that the tray **215** can be placed within the tray **201**. It should be noted that although in the illustrated embodiments, the trays **201** and trays **215** are generally trapezoidal in shape, the trays **201** and trays **215** may be embodied by other shapes not illustrated herein, including but not limited to, cubed, rectangular, cylindrical, and/or organic or ameba shapes. To this extent, both the tray **201** and the tray **215** may have additional or excluded boundaries (e.g., inner walls, outer walls, sides walls) than what is contemplated herein. As shown in FIG. 13D and FIG. 13E, in some embodiments, both the tray **201** and the pole **301** may be rectangular and/or cubed in shape such that the tray **201** and the pole **301** comprise four (4) side walls wherein at least one side wall of the tray **201** is adjacent to at least one side wall of the pole **301**. Alternatively, the tray **201** and the tray **215** may be circular in shape such that they comprise a single continuous perimeter wall. In such an embodiment, the grow tray may receive water from the pole assembly at a first portion of the perimeter wall, and transfer the water to and/or towards a second portion of the perimeter wall.

[0118] When placed in the tray **201** at least a portion of the base of the tray **215** rests adjacent to the side rails **209** of the tray **201** such that the side rails **209** support and uphold the tray **215** after it has been placed within the tray. Furthermore, in a preferred embodiment, the length of the tray **215**, as defined by the distance between the tray's inner wall **224** to its outer wall **225**, is less than the length of the tray **201**, as defined by the distance between its inner wall to the tray's outer wall **212**, **213**, such that the tray **215** is configured to slide along the side rails **209a**, **209b** back and forth between the inner and outer walls **213** and **214** of the tray **201**. In this regard, the first member **216** of the tray **215** may have at least one handle **222** that a user of the hydroponics system may grip to aid in sliding the tray **215** back and forth within the tray **201**. In one embodiment, as shown in the figures, the at least one handle **222** may be located at the top edge of the outer wall **225** of the tray **215**. When positioned on top of the side rails **209a**, **209b** towards the front of the tray **201**, the inner

wall **224** of the tray **215** may be adjacent to the inner wall **212** of the tray **201**. When positioned on top of the side rails **209a**, **209b** towards the back of the tray **201**, the outer wall **225** of the tray **215** may be adjacent to the outer wall **213** of the tray **201**. The tray **215** may additionally be positioned, on top of the side rails **209**, in any area between the inner and outer walls of the tray **201**. In one embodiment, the top of the tray **215** may extend beyond the top of the tray **201** after being placed on the rails **209a**, **26b**. In another embodiment, the top of the tray **215** may be flush with the top of the tray **201** after being placed on the side rails **209a**, **209b**. In still other embodiments, the tray **215** may not sit within the tray **201**, and instead may be span the opening of the tray **201**. As will be explained in further detail later the tray may be moved from a first position near the spout **400** to collect water in the channels **218**, **220** or may be moved to a second position so instead of delivering water to tray **215** the water is delivered to the tray **201** (e.g., when the roots grow from the tray **215** into the tray **201**). In yet another embodiment, the tray **215** may be positioned within the tray **201** in a third position such that the water is simultaneously dispensed into both the tray **215** and the tray **201** from the pole assembly. In such an embodiment the spout **400** may be embodied by a split configuration such that it directs the flow of the water to multiple locations, where a portion of the water is dispensed into the tray **215** and a portion of water is simultaneously dispensed to the tray **201**. Alternatively, in such an embodiment, the tray **215** may be embodied by a split configuration such that it directs the flow of the water to multiple locations, where a portion of the water is dispensed into the tray **215** and flows throughout the tray **215**, and a portion of water is immediately dispensed from the tray **215** to the tray **201**, prior to flowing throughout the tray **215**. In a preferred embodiment, the tray **215** is separable from the tray **201** to allow the roots to grow into the protrusions **202** of the tray **201**. In such an embodiment, the tray **215** may be removed from its position above the side rails **209** of the tray and further disassembled such that the first and second members **216**, **217** are separated from one another and the tray is removed from the hydroponics system **1** without harming the roots of the plants.

[0119] In one embodiment of the invention the tray **215** is located parallel with respect to the ground such that water only flows out of the tray when the water overflows the channel **218**, **220** walls. In one embodiment, side rails **209a**, **209b** in the tray **201** may be sloped towards the outer wall **213** of the tray **201** such that when the tray **215** is positioned on top of the side rails **209** the tray **215** is similarly sloped towards the outer wall **213** of the tray allowing the water to flow from the inner wall **224** of the tray **215** to the outer wall **225** of the tray **215** and overflow into the tray **201** below through a gap that is created between the outer wall **225** of the tray **215** and the outer wall **213** of the tray **201**. The gap may allow water to flow off the top of the tray and into the tray **201**. The tray **201** may then be sloped towards the pole assembly **300** such that after the water enters the tray **201** the water flows towards the pole assembly **300** and exits the tray **201** through one or more openings **206** in the inner wall of the tray **201**.

[0120] FIGS. **28** through **30** illustrate a spout **400** in accordance with embodiments of the invention. As illustrated in FIG. **1**, water may be transported from the pole assembly **300** to the hydroponics assembly **200** using a spout **400**. As illustrated in FIGS. **28** through **29**, the spout may be defined by a planar surface **401** that extends upwards from one or more spout hooks **402**. The spout **400** may be hooked onto the pole cup **305** and/or the tray **201**, such that the spout **400** at least partially rest on the edge of the upper tier **306** of the pole cup **305**, or the top of the tray hooks **208**. As shown in FIG. **1**, in one embodiment, the tray may have three (3) hooks **208** that are equally spaced apart such that spaces are formed in between the tray hooks **208**. The spout **400** may then have two (2) hooks **402** that securely fit within the spaces formed in between the tray hooks **208** such that the spout is coupled with both the edge of the pole cup **305** and the tray **201** when the hooks **402** are positioned within these spaces of the tray hooks

[0121] **208**. The planar surface **401** may then extend upwardly and inwardly from the inner wall of the tray **212** towards the pole **301** and/or exterior of the pole **301**, such that the water flows from the pole assembly **300** to the spout **400** and subsequently into the hydroponics system **20**. The top

edge of the spout **400** may be adjacent to either the pole **301** and/or exterior of the lower tier **307** of the pole cup **305**. The bottom edge of the spout **400** may be adjacent to the top of the inner wall of the tray **201**. In one embodiment, as shown in the illustrations, at least a portion of the sides of the spout **400** may be elevated such that the elevated portion concentrates the flow of water towards the center of the spout **400**. In some embodiments, sides may extend the length of the spout **400** in order to prevent or reduce the amount of water that flows over the sides of the spout **400** and never makes it to the trays **201**. In one embodiment, the sizes of the spouts **400** may be varied in order to supply different amounts of water to the trays **201**. In some embodiments the spouts **400** may cover the circumference of the pole **301** up to at least $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$, $\frac{1}{6}$, $\frac{1}{8}$, $\frac{1}{16}$, $\frac{1}{32}$, or the like of the circumference of the pole **301**. It should be understood that the various spouts **400** may cover any range of the circumference of the pole **301** that falls within, outside of, or overlap any range of circumferences of the pole **301** for these values. Spouts **400** of different sizes may be added or removed from the hydroponics system **1** in order to change the amount of water directed to one or more of the trays **201**.

[0122] In one embodiment, as shown in FIG. **1**, spouts **400** may be used to deliver water to the hydroponics assemblies **200**. In such an embodiment, water may either fill up in the pole **301**, or may be transported throughout the pole **301** and to the top of the pole **301** using an optional tube such that the water flows over the top edge of the pole or through one or more exit slots **313** in the pole assembly **300** where it is subsequently delivered to the hydroponics assemblies **200** via the spouts **400**.

[0123] In an alternative embodiment, as shown in FIG. **30**, puncture spouts **400** may be used to deliver water to the hydroponics assemblies **200**. In such an embodiment, the pole **301** may have a pole cap **312** installed that allows water to be released back into the tank **101** as pressure builds within the pole **301**. Additionally and/or alternatively a spout **400** may be placed directly into the pole **301** such that it penetrates the outer layers of the pole **301** and allows water to enter into the spout **400** directly from the pole **301**. In such an embodiment, the spout **400** may additionally comprise a flow rate control valve that varies the rate of flow of the water. For example, the flow rate control valve may be a ball or gate valve that includes a hard flap to divert the water down the spout **400** towards the tray **201**. The spout **400** may then spread water into the tray **201** with a lower portion of the spout **400** that may be wider or narrower than the rest of the spout **400**. In one embodiment, where the spout **400** is directly coupled with the pole **301** such that it penetrates the outer layers of the pole **301**, the pole **301** may be formed of a material that has rescalable properties such that the pole reseals itself after the spout **400** has been removed. In another embodiment, where the spout **400** is directly coupled with the pole **301** such that it penetrates the outer layers of the pole **301**, the pole **301** may have a plurality of pre-punctured holes configured to receive the spout **400** such that the pre-punctured holes may be covered with a cap when the spout **400** is not in use. This method may limit access to the pre-punctured hole when the spout is not in use thus restricting the continued flow of water from the pre-punctured hole.

[0124] It should be understood that the size of the hydroponic system **1** may vary depending on the end use of the hydroponic system **1**. As such, in one embodiment the hydroponic system **1** may be configured to fit on a countertop in a kitchen, school laboratory, or the like. In other embodiments, the hydroponic system **1** may be sized to fit in garages, greenhouses, or other like areas for a greater yield on the production rate of the vegetation. In still other embodiments of the invention, the hydroponic system **1** may be sized to fit within warehouses in which vegetation may be produced on a production level scale. In all of these systems different sizes of an aquaculture system (e.g., tanks **101**) may be utilized with multiple pole assemblies **300** and multiple hydroponic assemblies **200** on an as needed basis in order to grow the desired amount of vegetation. For example, two poles **301** may be stacked on top of each other in order to extend the height of the hydroponic system **1** for growing more food. In other embodiments more hydroponics assemblies **200** are added to the pole **301** to grow more food. Any number of pole assemblies **300** and/or

hydroponics assemblies **200** may be utilized to grow the desired amount of vegetation (e.g., food). [0125] FIG. **31** illustrates another example embodiment of a tray **410** for a hydroponic system as described herein, with water entering the tray along ramp **424** from the pole assembly (described further below) and flowing generally in the direction of arrows **440** across a planting area **415**. As shown, the illustrated embodiment of the tray **410** includes an outer periphery wall **412** that bounds the tray **410** at least along a majority of the periphery. This periphery wall **412**, in combination with inner walls **406a** and **406b** define channels **405a** and **405b**, respectively. A bottom of the tray **410** defines thereon a first set of projections **414** and a second set of projections **416**. The bottom of the tray **410** further defines ribs **418**. The first set of projections **414** can serve several functions. For example, the first set of projections **414** can retain a seed mat of a wicking material in a defined position within the tray **410**. The seed mat can be elevated with respect to the bottom of the tray **410** with the second set of projections **416** and/or the ribs **418**. Optionally, the first set of projections **414** can support a tray insert for the tray such as tray **215** of FIG. **25**. The first set of projections **414** and the second set of projections **416** allow water to flow through them while providing support and/or retention within the tray **410**.

[0126] According to the illustrated embodiment, the ribs **418** provide a low barrier for water as it flows through the tray. The ribs **418** promote a build up of water in front of the rib before allowing the water to flow over the rib. This precludes water from traveling through the tray **410** too quickly to be beneficial to plants growing in the hydroponic system. The ribs **418** further promote turbulent flow within the tray **410** which slows the flow of water along the bottom of the tray and reduces the surface tension of the water within the tray, thereby rendering the water more readily available to interface with roots of plants growing above the bottom of the tray. Laminar flow along the bottom of the tray may enable the water to flow in discrete channels or paths due to surface tension, which may allow the water to more easily avoid roots within the tray. Thus, the ribs **418** provide an improvement through both pooling of water and inducing turbulent flow to the water as it flows through the tray. Also illustrated in FIG. **31** are flow control features **420a** and **420b** which will be described in greater detail below.

[0127] The water flowing through the tray, directionally illustrated generally by arrows **440**, flows from the pole assembly **300**, along ramp **424**, and into the tray **410**. The water reaches a back wall of the periphery wall **412** after it flows over the ribs, through the second set of projections **416**, and through the first set of projections **414**. FIG. **32** illustrates a top view of the tray **410** of FIG. **31** with water flow depicted. Water flows into the tray **410** along ramp **424** from the pole assembly **300** and flows along arrows **440** across the planting area **415**, through the second set of projections **416**, over the ribs **418**, and through the first set of projections **414**. The water passes through the first set of projections **414** to a back channel **406**. The water flows from the back channel **406** to channels **405a** and **405b**, and back toward the pole assembly **300** shown by arrows **444**.

[0128] FIG. **33** illustrates a detail view of the mechanism by which the tray **410** attaches to the pole assembly **300** and the flow path of the water to and from the pole assembly. As shown, the attachment mechanism includes a pair of protrusions **430**, each with a corresponding recess **432**. Between the pair of protrusions is ramp **424** that receives water from the pole assembly **300**. Water traveling down the outer surface of the pole assembly reaches the ramp which, in the assembled position, engages the pole assembly about a portion of a periphery of the pole assembly such that water flowing down along the outer surface of the pole assembly meets the ramp **424** and flows into the tray **410** along the path shown by arrow **446**. The water flows through the tray as illustrated and described with respect to FIG. **32** above, and returns to the pole assembly **300** via channels **405a**, **405b** through spout **426** shown in FIG. **33** along a flow path shown by arrow **448**. Spout **426** is supported away from the pole assembly **300** by tray supports **422** that provide a gap between the spout **426** and pole assembly **300** through which water is returned to the outer wall of the pole assembly.

[0129] FIG. **34** illustrates a detail view of the flow control feature **420b** illustrated in FIG. **31**. The

flow control feature **420b** is a feature that is used to control the flow of water entering channel **405b**, thereby regulating the depth of water inside the tray **410**. The flow control feature **420b** can be implemented in a number of ways, one of which is illustrated in FIG. **34** which depicts slots **421b** on either side of the flow path leading to channel **405b**. FIG. **35** illustrates a barrier **422b** inserted into slots **421b** forming the mechanism for flow control. The barrier can be inserted fully such that it is seated against a bottom of the tray, in which case it functions as a dam, requiring water to build up behind the barrier and permit flow to the channel **405b** only when the barrier is over-topped. Such a configuration would lead to a pooling of water within the tray **410** to a height at least equal to the height of a top of the barrier **422b**. This can allow roots of plants growing in the hydroponic system to remain in water as may be necessary for certain hydroponic configurations.

[0130] Optionally, the barrier **422b** can be elevated with respect to the bottom of the tray **410**. When the barrier **422b** is elevated relative to the bottom of the tray **410**, water will continue to flow under the barrier through a restricted flow path. Pressure will build on the water flowing under the barrier as the water level rises in the tray **410**, provided the barrier is at a position in which the level of water flow into the tray is greater than the flow under the barrier (without pooling). The increased pressure will drive more water through the area under the barrier, which will reach a point of equilibrium between a pooling height of water within the tray **410**, and the flow of water pressurized by the pool height under the barrier. Such a configuration allows control of the pooling level within the tray **410**. Advantages to the flow control feature **420b** and the resultant pooling within the tray **410** will be described further below with respect to different tray and tray configurations for the hydroponic system.

[0131] According to some embodiments, the barrier **422b** may be formed of a porous material that allows water to flow through the barrier, but precludes the flow of seeds or small plants should they be carried by the water flow to the flow control feature **420b**. Such a porous material can include, for example, a screen such as a porous version of the barrier **422b**, with holes sized based on the desired flow rate through the barrier and based on a size of object to be screened from passing through the water control feature. The holes through such a screen may be large enough to permit sufficient water flow, but small enough to preclude the flow of seeds through the screen. Further, the holes may be shaped and sized such that seeds or small plants do not block the holes. Optionally, the screening material may be in the form of a woven or non-woven material. Examples include a sparse unwoven polymer such as a cellulose, nylon or spun polypropylene fiber. Such a material may not be seated within the slots **421b**, but may be pressed into the region of the flow control feature as such materials are pliable. These non-woven polymer materials can be beneficial as they will permit the flow of water through them, while precluding the flow of solids such as seeds or small plants and the material will not become plugged or blocked by such solids, thereby reducing maintenance and potential modes of failure.

[0132] FIG. **36** illustrates another example embodiment of a tray **810** as described herein. The above-described trays **410** include a circular section shape, where a plurality of trays, such as five trays, arranged about a pole cup form a full circle of circular section shaped trays **410**. Further embodiments include substantially square-shaped or rectangular-shaped trays as shown in FIG. **36**. As shown, the tray **810** of the illustrated embodiment includes a square shape with a curvature in the corner **813** to facilitate attachment to a pole cup as described in the aforementioned embodiments. The attachment mechanism to the pole cup may be the same as described above, such that further description will not be provided herein with reference to the square tray **810**. At the corner **813**, the tray includes a ramp **825** similar to ramp **824** described above. This ramp **825** guides water from the pole section into the tray, along entry channel **827** as shown by arrow **823** when the tray **810** is in an installed position relative to a pole cup. The entry channel **827** is separated from the planting area **815** of the tray **810** by a sawtooth wall **829**. This sawtooth wall **829** functions to allow water to build within the entry channel **827** to some degree before spilling

through between the teeth of the sawtooth wall. This construction enables substantially equally distributed flow into the planting area **815** of the tray **810** along the length of the sawtooth wall **829**. The water flows across the planting area **815**, through protrusions **814**, and into return channel **837** along arrows **833**.

[0133] The tray **810** is configured such that in the installed position relative to a pole cup the water received into return channel **837** flows through a gateway serving as a water control feature **835**. This water control feature **835** may be dammed as described above with respect to FIG. **36**. Water flowing under such a dam, over such a dam, or through the gateway flows into exit channel **840** along arrow **839** to return to the pole cup, exiting the tray **810** along a path similar to that illustrated with respect to FIG. **33** above. The square tray **810** of example embodiments can increase a planting surface area in some embodiments, with four such squared trays installed in a single pole cup, forming a rectangular or square tray area at each level of the tower of pole cups.

[0134] Different types of plants benefit from different structures for optimal growth within the trays described herein. Seeds of plants can be retained by the protrusions described above where water flows over the seeds for germination. Optionally, seeds may be received on a seed mat which includes a wicking material that wicks material flowing through the tray to keep seeds moist for germination. Another example embodiment includes seeds or seedlings retained in pods that facilitate growth of the plants. Such an embodiment can include a structure configured to receive pods and suspend them at a predetermined height relative to a water level within a tray. FIG. **37** illustrates an example embodiment of a tray lid **434** configured to receive therein pods into respective pod apertures **436**. The tray lid **434** of the illustrated embodiment is configured to attach to a tray **410**, where the tray may be configured in any of the aforementioned configurations. The pod apertures **436** may be sized to receive therein pods, where the pods are vessels into which seeds or seedlings are received. The pods of example embodiments may be formed of a wicking, absorbent material, such as cotton, man made fibers, coconut fibers, paper, or the like.

[0135] FIG. **38** illustrates a section view schematic of a single pod aperture **436** within the tray lid **434** including a pod **438**. As shown, a seedling **439** is received within the pod **438**, and the pod rests within the pod aperture **436** with a flange of the pod supporting the pod relative to the tray lid **434**. Also shown is a bottom surface **411** of a tray and a water level **435**. The pod **438** can wick the water to keep the seedling **439** moist to facilitate growth. The water level **435** can be adjustable using the flow restricting embodiments described above with respect to FIGS. **34** and **35** to ensure the height of the water level is appropriate for the pod **438** received within the pod aperture **436**. The illustrated example embodiment of FIGS. **37** and **38** may benefit specific types of plants, particularly those that require adequate spacing to grow well, such as a lettuce where a head of lettuce may require spacing around the plant as it grows to optimize the growth and obtain maximum benefit from the systems described herein. The pod aperture size and arrangement of the tray lid **434** shown in the example embodiment can be varied to accommodate different size pods and different spacing depending upon the type of plant to be grown. A height of a surface tray lid **434** can be configured based upon the type of plant to be grown therein together with the type and size of pod to be received within the respective pod apertures **436**. The tray lid **434** of an example embodiment can employ spacers between the tray and the tray lid to appropriately distance the surface of the tray lid from the tray to properly accommodate the plants and/or pods received in the tray lid.

[0136] The pods received within pod apertures **436** of the embodiments described herein can be supplied with water via damming of the water in the tray to raise the water level to that of the pod as shown in FIG. **38**. However, embodiments described herein do not require pooling of the water in a tray to reach the pods. According to an example embodiment, the pods within a tray lid **434** may be fed by water guided specifically to a respective pod aperture **436**. FIG. **39** illustrates an example tray **410** including flow directing fingers **425** that are configured to guide water from the pole section (proximate the ramp **424**) to the individual pod locations, shown by pod apertures **436**.

The water is guided along arrows **427** to each individual pod, thereby wetting the pod before the remaining water falls into the tray, and exits along the arrows as shown in FIG. **32** described above. This configuration eliminates the need for pooling of the water within the tray **410**. Optionally, each of the flow directing fingers **425** may terminate beneath a respective pod aperture and include a small pool to hold water directly below a pod aperture to ensure the respective pod absorbs sufficient water. The elimination of the need for pooling within the tray **410** reduces the weight of the tray during operation and reduces total water held by a system, thereby cycling water more quickly in its entirety. Such water cycling may be important in balancing the nutrients in the water, particularly in an aquaponics embodiment.

[0137] The flow directing fingers **425** are elevated with respect to a floor of the tray **410**, but at an angle such that water traveling from the pole section **456** will travel along each flow directing finger and eventually flow into the tray **410** to be returned to the pole cups. FIG. **40** illustrates a profile view of a configuration in which the water travels along a flow directing finger **425** to a pod **438** visible both above the tray lid **434** and below. Each aperture **436** may have a corresponding flow directing finger **425**. The flow directing fingers **425** may be coupled together and be supported within the tray **410** by structures therein, such as the protrusions, channel walls, or the like. Such support can hold the flow directing fingers **425** fixed position relative to the tray lid **434** and thus the pod apertures **436**.

[0138] FIG. **41** illustrates another example embodiment of a tray **410** with a mechanism for supplying water to pods located within pod apertures **436** shown in FIG. **37**. The illustrated embodiment of FIG. **41** includes a trough **423** where water collects in a center pool of the trough **423**, fills the pool, and subsequently flows out through channels **428** of the trough. The trough **423**, much like the flow directing fingers **425**, are configured based on the pod apertures **436** within the tray lid, such that the channels **428** of the trough **423** carry water to each pod. In some embodiments, a channel may supply water to more than one pod, or a single pod may be supplied by multiple channels. The ends of each channel are, in some embodiments, configured to contact or be in close proximity to a pod received within a respective pod aperture **436**. This ensures water flowing through each channel reaches a respective pod. The pool within the trough **423** serves to provide even distribution of the water among the different channels **428** of the trough **423**, while also slowing the velocity of the water as it flows into the pool before being distributed through the channels.

[0139] FIG. **42** illustrates a detail view of where the tray **410** meets the pole cup with protrusions **430** that engage the bottom edge **433** of the upper cup **452** as described above. The trough **423** includes a trough ramp **424A**, configured to overlies the ramp **424**, such that water that would be received along ramp **424** is instead received at trough ramp **424A** and carried to the pool of the trough **423** shown in FIG. **41**. Further, as depicted in FIG. **42**, the trough **423** includes trough tabs **423A** that engage inner walls **406A** and **406B** to support the trough **423** within the tray **410**. These trough tabs **423A** provide sufficient support to suspend the trough **423** above the bottom of the tray **410**. Further, the trough **423** includes trough recesses **432A** that function similar to recesses **432** to engage a bottom edge **433** of the upper cup **452** as described above with respect to FIG. **33**. The trough recesses **432A**, which are raised relative to trough ramp **424A**, serve to hold down the trough even with the weight of the water carried through the trough.

[0140] Water distribution within the trays of example embodiments described herein is important to ensure proper supply of water to each of the plants within a tray. While the water flow directing fingers **425** and the trough **423** described above provide water flow to pods and the like, certain trays are configured for supplying water to seeds positioned directly onto the tray, or onto an insert of the tray. Such a configuration may be beneficial for the growing of certain plants, such as micro greens, mung beans, or the like. FIG. **43** illustrates an example embodiment of a tray insert **802** that is configured to facilitate water flow within a tray. The illustrated embodiment of FIG. **43** includes a top view, bottom perspective view, side view, and top perspective view. The tray insert **802** is

configured to be received within trays **410** described above and to promote different flow characteristics of water within the tray that may be better suited to different plants without requiring changing of a complete tray. The top view of the tray insert **802** illustrates the wavy flow guides **804** extending radially outward from the front **806** of the tray insert. These wavy flow guides **804** function to slow the speed of water as it flows from the front **806** of the tray insert to a back **808** of the tray insert. Further, the wavy flow guides **804** contribute to turbulence of the flow of the water, which both slows and aerates the water to some degree, while increasing the dwell period of water along the surface of the tray insert **802**.

[0141] The tray insert **802** of FIG. **43** further includes ribs **811** extending concentrically across a width of the tray. These ribs **811** function to introduce turbulence of the water flowing through the tray, increase dwell time of water along the tray as water builds in front of the ribs, and aids with even distribution of water across the width of the tray insert. The tray insert **802** of FIG. **43** has both a top side and a bottom side, and can be used with either the bottom side or the top side facing up. The bottom side differs from the top in that the bottom side includes channels **812**. These channels **812** provide for drainage from the tray insert **802** to help preclude over saturation of plants within the tray insert, but also serve in a manner similar to that of the ribs **811** by increasing turbulence, increasing water dwell time, and promoting even distribution of water across a width of the tray insert.

[0142] FIGS. **44-47** illustrate another example embodiment of a modular system for pole cups. FIG. **44** illustrates a side-view of a single pole cup **450** module, where a plurality of pole cups are configured to be assembled into a vertically arranged pole. The pole cup **450** of FIG. **44** includes an upper cup **452** and a collector **454** connected by a pole section **456**. The upper pole cup defines a lower pool **458** whose depth is defined based on the position of the lower pool base **460** within the pole section **456**. Above the lower pool **458** and separating the upper cup **452** from the pole section **456** are a plurality of blades **462**, between which are open spaces through which water can flow from the upper cup down along the pole section. Through the middle of the pole cup **450** is disposed a tube **464**. At the base **466** of the collector **454** are collector blades **468**, which separate the collector base **466** from the pole section **456**.

[0143] FIG. **45** illustrates two pole cups **450** assembled together, where the base **466** of a first pole cup is received within the upper cup **452** of a second pole cup, and the tube **464** of the first pole cup engages and seals against the tube **464** of the second pole cup to form a continuous tube there through. The pole cups cooperate to form a flow path for water as it flows through the system of example embodiments described herein. The flow path of the water through the pole cups of the system is illustrated in FIG. **45** with arrows. As shown, water rises through the tube **464** and exits through the tube of the uppermost pole cup in the system into upper cup **452** along arrow **470**. The water collects in the lower pool **458** of the pole cup **450** and the lower pool overflows through blades **462**. The water flowing between the blades **462** flows along the outer wall of the pole section **456**. In practice, a tray, such as the tray illustrated in FIG. **31** is attached to the pole cup **450** and supported by collector **454** with protrusions **430** (shown in FIG. **33**) engaging the bottom of the upper cup **452**, described further below. When a tray is attached to the pole cup system, water flows out of the upper cup **452** between blades **468** down along the pole section **456** until it encounters the ramp **424** of the tray (illustrated in FIG. **33**), where the water is diverted into the tray flowing in a direction along arrow **472**. This portion of the flow path corresponds with path shown by arrow **446** in FIG. **33**. Water is returned from the tray along arrow **474** of FIG. **45**, corresponding with the flow path illustrated by arrow **448** in FIG. **33**.

[0144] The water returned from the tray along arrow **474** is received by the collector **454**. The water flows through the collector **454** and between collector blades **468** and into the upper cup **452** of the next pole cup **450** down. This flow path is repeated for each pole cup **450** in the assembled tower, such as shown at arrow **476** of the subsequent pole cup receiving water through the upper cup **452** to a next tray, and returned to the collector from the tray along arrow **478** to continue the

flow path.

[0145] The distance between pole cups can be varied using different sizes of pole cups, or alternatively, spacers in the form of pole sections that extend between the base **466** of one pole cup to an upper cup **452** of a next pole cup. In such a spacer, the water would flow from inside a collector **454**, through the collector blades **468** and through the inside of the spacer into the next lower upper cup **452** of the assembled system. Such spacing allows for taller plants as trays are separated by a greater distance. Optionally, trays can further be axially spaced relative to one another without the need for spacers by omitting trays from a pole cup, and allowing a pole cup to let water pass there through, from the blades **462** of the upper cup **452**, along the wall of the pole section **456**, and into the collector **454** without reaching a tray **410**. This same bypass occurs when a tray **410** is removed from a pole cup **450** for harvesting, service, or otherwise moved. As such, removal and insertion of trays **410** can be performed while water flows through the system without interfering with the water flow through other trays.

[0146] According to an example embodiment, a lid or cap may be attached to an uppermost pole cup **450** atop the upper cup **452** such that water flowing up through the tube **464** is retained within the upper cup and is diverted along the appropriate flow path. According to further embodiments described herein, the water may be received within the upper cup **452** not through the tube **464**, but from another water source that functions in the same manner to provide water to the upper cup with sufficient flow volume to feed each of the trays of an assembled system.

[0147] FIG. **46** illustrates the attachment process for attaching a tray **410** to a pole cup **450** as described herein. The tray **410** is tilted relative to the pole cup **450** in order to advance protrusions **430** below a bottom edge **433** of the upper cup **452**. The tray **410** is then move to a position extending perpendicular to an axis through the pole cup **450** as shown in FIG. **47**. This movement causes the protrusions **430** to engage the bottom edge **433** of the upper cup **452**, while the bottom edge **433** of the upper cup **452** is received within recesses **432**. The tray **410** rests on the collector **454** in the installed and seated position shown in FIG. **47**. In the installed and seated position, an upper end of the ramp **424** (shown in FIG. **33**) is pressed into contact with the pole section **456**, which enables the water flowing down along a surface of the pole section to be deflected by the ramp **424** to enter the tray. The tray supports **422** engage the pole section **456** to stabilize and support the tray **410** against the pole section. Spout **426** allows water to flow from the tray **410** back to the pole section **456** and into the collector **454**. This attachment configuration ensures a firm engagement between the tray **410** and the pole cup **450** to ensure the proper flow of water and to provide structural rigidity for the system described herein.

[0148] FIG. **48** illustrates the tray **410** fully seated and engaged with the pole cup **450**. As shown, the tray rests on collector **454** while protrusions of the tray are engaged within the bottom edge **433** of the upper cup **452**. This arrangement provides a secure coupling between the tray **410** and the pole cup **450**, and can support weight within the tray as the tray will hold some amount of water and the plants growing therein. FIG. **49** illustrates five trays **410** arranged around a single pole cup **450**. As shown, the trays **410** function independently and can be independently removed and attached to the pole cup. Further, the trays can each be configured differently from one another (e.g., with lids, inserts, etc.) as each tray **410** functions independently. Connecting pole cups **450** together as described above with respect to FIG. **45** results in a stack or tower of pole cups, each able to receive a plurality of trays as illustrated in FIG. **49**. This arrangement provides a compact and efficient arrangement of growing trays to produce a relatively large amount of plants within a relatively small footprint.

[0149] FIG. **6** above illustrated an example embodiment of a substantially square tray. FIG. **50** illustrates a top view of four of these trays **810** as secured to a pole cup **450**. The substantially square trays **810** of FIG. **50** are illustrated with tray lids **834** including pod apertures **836**. As shown, the square form factor of the illustrated trays **810** provide a larger surface area for planting, which can accommodate more pods. However, the square form factors of these trays preclude

rotation of a tower of pole cups and trays if one side of the tower abuts a surface. The circular form factor of the trays shown in FIG. 50 can be rotated about an axis of the pole cup even when a tower of such trays are arranged to abut a surface or adjacent to another tower of pole cups and trays. FIG. 51 illustrates a perspective view of the square tray arrangement of FIG. 50.

[0150] The system of example embodiments described herein can be used in a variety of implementations. As a compact form factor, several pole cups can be used for one or more layers of trays as a tabletop version, similar to that depicted in FIG. 1. In such an embodiment, the tank 101 can provide the water source, while the water is pumped through tube 464 to the top of the assembled pole cups before descending down through the pole cups 450 and any attached trays 410. Optionally, embodiments can be assembled into larger systems, such as a tower several feet tall.

[0151] One such example embodiment of a system described herein includes a mobile implementation 480 as depicted in FIG. 52. As shown, a tower of pole cups 450 are assembled in a vertical arrangement with a plurality of trays 410 attached thereto. Further included is a tank 482 mounted on mobile platform 484. The mobile implementation 480 can be moved as needed. The mobile implementation includes an upright member 486 which functions as a structural member to secure the top of the tower of pole cups 450 to prevent swaying and to ensure structural satiability of the mobile implementation 480. The structural member can be mounted to the mobile platform 484 or to the tank 481 as needed, and can include structural supports 488. The manner of operation of the mobile implementation 480 can take two forms. A first includes where the water is pumped through the tube within the pole cups to reach the top of the tower of pole cups where the water begins the descent that feeds each tray. Optionally, the water may be fed through a hose or pipe in the structural member 486 to reach the top of the pole cup tower. In either manner of operation, water is pumped to the top of the pole cup tower before being recouped at the bottom of the tower, which is in the tank 482 in the illustrated embodiment of FIG. 52.

[0152] In addition to a mobile implementation of a single tower of pole cups, embodiments described herein further include a mobile array of towers of pole cups. Such a mobile array may be in the form of a plurality of towers of pole cups arranged within a shipping container and fixedly positioned within the shipping container. FIG. 53 illustrates an example embodiment of a defined space including a plurality of towers of pole cups in fixed positions. The example implementation shown in FIG. 53 can be implemented within a shipping container, or implemented in a space such as a grow house or other location where growing operations take place. One significant benefit of the system illustrated in FIG. 53 is the ability of the system to be self-contained such that it can be implemented in a shipping container and deployed to any destination that can benefit from the mobile grow room. FIG. 53 illustrates one such structural arrangement; however, it is understood that the components described herein can be arranged in different configurations while obtaining the same benefits of the system describe with respect to FIG. 53.

[0153] FIG. 53 is a side view of an arrangement of towers of pole cups arranged along walls, such as in a shipping container or otherwise confined space. As shown, the towers 700 disposed on three walls of the container each include a plurality of levels of trays 710 arranged around the tower of pole cups. The towers 700 are supported at the bottom by a support bracket 712 shown in FIG. 54 which depicts the bottom of an array of towers 700. The bracket provides a support for the weight of the tower of pole cups 708, with a base of a bottom pole cup 708 received within support bracket cup 714. The support bracket 712 with the support bracket cup 714 support the weight of the tower of pole cups and levels of trays 710. The support bracket 712 can be mounted to a structural frame or to a wall of a shipping container (either directly or indirectly) and is shown cantilevered from a structural frame member 716, which may be secured to a wall of a shipping container in a containerized embodiment. The support bracket 712 thus suspends the tower of pole cups above a floor of the space or shipping container. The support bracket 712 can optionally include a support extending to a floor to aid in supporting the weight of the tower of pole cups and levels of trays.

[0154] Also illustrated in FIG. 54 is a conduit 718 from the support bracket cup 714 leading to drain manifold 720. Water flowing through the base of pole cup 708 flows into the support bracket cup 714, and drains from the support bracket cup through conduit 718 into the drain manifold 720. This occurs for a plurality of towers 700 and potentially all towers within the system of the confined space or container. There could be more than one drain manifold 720 to accommodate different layouts; however, all towers 700 within the system of the confined space can drain through respective conduits (e.g., conduit 718) to a common drain manifold 720. The drain manifold 720 conducts the water drained from the towers into a tank 722. This tank functions in a similar manner to the tanks of above-described embodiments; however, tank 722 collects water discharged from a plurality of towers. To that end, the tank 722 can operate as a tank in a hydroponic operation or an aquaponic operation supporting aquatic life.

[0155] The tank 722 is also used to supply the towers 700, at least indirectly as detailed further below, with water, such that the water system of the hydroponic or aquaponic system within the confined space or container is closed loop or substantially closed loop. Substantially closed-loop describes an embodiment in which water may be added to the system to compensate for water consumed and water lost through evaporation, leaks, spills, or the like. While the system is designed to operate as closed-loop, embodiments may be described as substantially closed loop where water and potentially other additives (e.g., water nutrients) are added to the system periodically to compensate for losses.

[0156] FIG. 55 illustrates a top side of the towers 700 within the confined space or container including an upper support bracket 730 with upper support bracket cup 732. The upper support bracket 730 is secured to the structural frame member 716, which may be the same structural frame member as the support bracket 712 of the same tower 700 is connected to. The upper cup of the top-most pole cup or an extension therefrom (e.g., a spacer) is received within the upper support bracket cup 732, and a supply conduit 734 is arranged to supply water from a supply manifold 736. The supply manifold 736 may be configured to supply all or some of the towers 700 within the confined space or container. The water supplied to the towers cascade through the pole cups and the trays as described above.

[0157] FIG. 56 illustrates an example embodiment of a pump system for circulating water through a plurality of towers according to an example embodiment. As illustrated, the drain manifold 720 conducts water discharged from the towers 700 to tank 722. The tank 722 of the embodiment of FIG. 56 is an intermediate tank, with a pump 726, such as a submerged sump pump arranged to pump water from the intermediate tank 722 to a primary tank 740 via return line 724. The primary tank 740 of the illustrated embodiment supplies water to primary pump 742 via inlet line 744. The primary pump 742 can be a conventional pool pump, which may include variable speed pumping capabilities and an integrated filter basket according to the illustrated example. The pump outputs the water along output line 746 to one or more supply manifolds 736, which in turn supply water to each of the towers as shown in FIG. 55. The illustrated embodiment of FIG. 56 depicts multiple supply manifolds 736, which may each be coupled to the output line 746 by a valve such. Each supply line among multiple supply lines may supply water to a different subset of the towers 700. Such a configuration may be beneficial to maintain zones within the confined space or container, where a zone can be individually shut off for plant harvesting, plant seeding, or maintenance of towers in that zone without shutting off water to all towers of the system.

[0158] FIG. 57 illustrates an example cutaway of a hydroponic system as described herein as arranged within a shipping container. The components described above are illustrated within a confined area demonstrating the compact and efficient system of example embodiments. The shipping container 790 can house all components of the hydroponic system to permit transport of an entire self-contained hydroponic system to any location. Further, besides the seeds or seedlings needed to produce plants, the only resources needed for such a containerized system is power and water. The water is largely recyclable in a substantially closed loop, though water topping off will

occur with prolonged operation. A freshwater tank can be employed, separate from primary tank **740**, where the freshwater tank can be used to supplement the water in the primary tank **740** that is circulated through the towers. Further, a drain line may be provided from the supply manifolds **736** to the primary tank **740**. A further line may be provided between the output line **746** and the drain manifold **720** proximate a distal end of the drain manifold to facilitate a drain line flush operation. One or more filters may be installed within the plumbing lines, such as at the outlet of the pump **742**, at the inlet of the pump, or elsewhere to remove particulates from the water that may adversely affect performance of the system described herein.

[0159] The plumbing configuration illustrated in FIGS. **54-57** is an example embodiment of a plumbing system for the hydroponic tower system described herein; however, various other plumbing configurations can be employed. For example, the drain manifold **720** can function as the water tank or reservoir draining directly to the pump **742**, with a supplemental water tank available to top off the drain manifold reservoir as needed.

[0160] The power can be provided by a local power grid, or alternatively through solar cells mounted to the container **790** to form a completely self-contained hydroponic growing system. The power is needed to run the water through pump **742** to supply each of the towers **700** of trays **710**. The pump may be a high-efficiency variable speed pump to minimize power consumption. Power may optionally be needed for grow lights that can be mounted to the container (e.g., to the ceiling) or to the towers **700**, such as mounted below trays **710** to supply light to each tray individually. The power provided to the shipping container **790** is optionally used for additional functions, such as climate control, dehumidification, computer devices (e.g. timers, controllers, etc.), communications system (e.g., WiFi, cellular communications, etc.), or the like.

[0161] According to some embodiments, the container **790** can be used as a hub for communications, particularly in areas lacking reliable communication methods. For example, the container **790** can include a satellite communications link, and can use this link to provide wireless communications for those proximate the container, such as through WiFi or other near-field communications protocols to connect people near the container with the rest of the world. The container of example embodiments can further be used to filter and provide clean drinking water to a community. While water from a water source (e.g., a well) may be suitable for growing plants, the water may not be sufficiently clean for human consumption. The water processed through the systems described herein can be cleaned by the plants to produce clean drinking water from the water returned from the trays. This system can clean ground water to provide clean drinking water to a community where the container is located.

[0162] According to some embodiments, grow lights may be disposed in asymmetrical locations with respect to the different levels of the towers, such that the towers may be configured to rotate to permit equal light distribution about each level of the tower. Rope lights or strips (e.g., strips of LED lights), pole elongated lights, panel lights, and light reflective panels or curtains can further be used to spread light across the planting areas of the trays to facilitate efficient growth. According to still other embodiments, windows or skylights may be positioned within a top of a shipping container to allow natural light to illuminate the towers and thereby facilitate growth. Such windows or skylights may be openable as needed.

[0163] The towers **700** described herein can implement interchangeable parts, where pole cups and trays (e.g., pole cup **450** and tray **410** or **810**) can be used in any tower at any location.

Interchangeable parts minimizes the variations that need to be manufactured and renders spare parts easily maintained, particularly when a containerized embodiment of the hydroponic system is remotely located. Interchangeable parts further enables the tower heights to be customizable without requiring new components. For example, if plants to be grown become tall, the trays from a level above the tall plants may be omitted to allow the plants to grow upward. Removal of the trays in the system described herein can be performed while water is flowing without disrupting the hydroponic system of example embodiments. Further, the upper support bracket **730** may be

movably engaged along a vertical structural frame member **716** to alter the overall height of each tower **700**.

[0164] While certain exemplary embodiments have been described and shown in the accompanying drawings, it is to be understood that such embodiments are merely illustrative of and not restrictive on the broad invention, and that this invention not be limited to the specific constructions and arrangements shown and described, since various other changes, combinations, omissions, modifications and substitutions, in addition to those set forth in the above paragraphs, are possible. Those skilled in the art will appreciate that various adaptations, modifications, and combinations of the just described embodiments can be configured without departing from the scope and spirit of the invention. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described herein.

[0165] Also, it will be understood that, where possible, any of the advantages, features, functions, devices, and/or operational aspects of any of the embodiments of the present invention described and/or contemplated herein may be included in any of the other embodiments of the present invention described and/or contemplated herein, and/or vice versa. In addition, where possible, any terms expressed in the singular form herein are meant to also include the plural form and/or vice versa, unless explicitly stated otherwise. Accordingly, the terms “a” and/or “an” shall mean “one or more,” even though the phrase “one or more” is also used herein

[0166] Many modifications and other embodiments of the disclosure set forth herein will come to mind to one skilled in the art to which these embodiments pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A tray for a hydroponics system comprising: a planting area within the tray; a ramp for guiding water from a water source into the planting area of the tray; at least one channel for returning water from the planting area of the tray to a spout exiting the tray; and at least one adjustable water flow control mechanism within the tray between the at least one channel and the planting area, wherein the at least one adjustable water flow control mechanism controls, in combination with a flow rate of water from the water source, a pooling depth of the water within the planting area of the tray, and wherein the pooling depth is adjusted in response to adjustment of the at least one adjustable water flow control mechanism.
2. The tray for a hydroponics system according to claim 1, wherein the at least one adjustable water flow control mechanism comprises a barrier, wherein the barrier is adjustably positioned between the planting area and the at least one channel.
3. The tray for a hydroponics system according to claim 2, wherein the barrier, in a first position, allows water to flow from the planting area, under the barrier, and into the at least one channel, and wherein the barrier in the first position causes pooling in the planting area.
4. The tray for a hydroponics system according to claim 3, wherein the barrier, in a second position, allows water to flow from the planting area, over barrier gate, and into the at least one channel, wherein the barrier in the second position causes pooling in the planting area at least to a height of a top of the barrier relative to a bottom of the planting area.
5. The tray for a hydroponics system according to claim 1, further comprising: at least one pair of slots between the at least one channel and the planting area, wherein the at least one adjustable water flow control mechanism comprises at least one barrier, wherein the at least one barrier is slidably received within the at least one pair of slots.

- 6.** The tray for a hydroponics system according to claim 1, further comprising a plurality of protrusions positioned between the planting area and the at least one channel, wherein the plurality of protrusions permit water flow between them.
- 7.** The tray for a hydroponics system according to claim 1, further comprising an entry channel, wherein water entering the tray is guided from the ramp to the entry channel before entering the planting area of the tray.
- 8.** The tray for a hydroponics system according to claim 7, further comprising a wall separating the entry channel from the planting area of the tray, wherein the wall is at least partially permeable by water to permit water flow from the entry channel into the planting area.
- 9.** The tray for a hydroponics system according to claim 7, wherein the tray comprises a substantially rectangular shape, wherein the entry channel is disposed along a first side of the tray, water flows across the planting area to a second side of the tray, opposite the first side of the tray, and water returns along the at least one channel for returning water from the planting area along a third side of the tray, wherein the ramp is disposed between the first side of the tray and the third side of the tray.
- 10.** The tray for a hydroponics system according to claim 1, further comprising a lid covering the planting area, wherein the lid defines at least one aperture for receiving therein a pod.
- 11.** The tray for a hydroponics system according to claim 10, further comprising a pod, supported in the at least one aperture, wherein the pod is configured to wick water from the planting area into the pod.
- 12.** The tray for a hydroponics system according to claim 11, further comprising flow directing fingers disposed between the lid and the planting area, wherein the flow directing fingers receive water from the water source and direct the water to pods received in each of the at least one aperture.
- 13.** A tray for a hydroponics system comprising: a planting area within the tray, wherein the planting area receives water from a pole to which the tray is attached; a channel for returning water from the planting area of the tray to a spout exiting the tray; and a water flow control mechanism within the tray between the channel and the planting area, wherein the water flow control mechanism at least partially controls a pooling depth of the water within the planting area of the tray.
- 14.** The tray for a hydroponics system according to claim 13, wherein the water flow control mechanism is adjustable, the tray further comprising: a pair of slots between the channel and the planting area, wherein the water flow control mechanism comprises a barrier, wherein the barrier is slidably received within the pair of slots, wherein the pooling depth is adjusted in response to adjustment of the water flow control mechanism.
- 15.** The tray for a hydroponics system according to claim 13, wherein the water flow control mechanism comprises a barrier, wherein the barrier is positioned between the planting area and the channel.
- 16.** The tray for a hydroponics system according to claim 15, wherein the barrier is adjustable, in a first position, allows water to flow from the planting area, under the barrier, and into the channel, and wherein the barrier in the first position causes pooling in the planting area.
- 17.** The tray for a hydroponics system according to claim 16, wherein the barrier, in a second position, allows water to flow from the planting area, over the barrier, and into the channel, wherein the barrier in the second position causes pooling in the planting area at least to a height of a top of the barrier relative to a bottom of the planting area.
- 18.** The tray for a hydroponics system according to claim 13, further comprising an entry channel, wherein water entering the tray is guided through the entry channel before entering the planting area of the tray.
- 19.** The tray for a hydroponics system according to claim 13, wherein the water flow control mechanism comprises a flow restriction between the planting area and the channel.

20. The tray for a hydroponics system according to claim 19, wherein the water flow control mechanism comprises a water-permeable barrier, wherein the water-permeable barrier is formed of a porous material.
